Demand Estimation with Market-Level Data: A BLP Approach with Optimal Instruments

Kenji Watanabe, Aisha Mensah, Dmitri Sokolov

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Abstract

We estimate a random coefficients logit demand model (BLP) using market-level scanner data from 342 geographic markets. Exploiting optimal instruments constructed from the expected Jacobian of market shares, we obtain price coefficients with mean -0.241 and standard deviation 0.118, implying mean own-price elasticities of -3.71. The implied Lerner markups closely match accounting-based estimates. Merger simulation predicts price increases of 6.8% for merging parties and 1.9% for rivals, with consumer surplus declining by \$3.12 per consumer per quarter. Monte Carlo results confirm that BLP with optimal instruments achieves near-nominal coverage at moderate sample sizes.

Contents

I Introduction

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on industrial organization by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in industrial organization is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides

a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

$$u_{ijt} = \alpha_i p_{jt} + x'_{jt}\beta_i + \xi_{jt} + \varepsilon_{ijt}, \quad \alpha_i \sim \mathcal{N}\left(\bar{\alpha}, \begin{bmatrix} \sigma_\alpha^2 & 0 \\ 0 & \sigma_\beta^2 \end{bmatrix}\right), \quad \binom{J}{2} \text{ cross-elasticities} \tag{1}$$

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in industrial organization since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

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A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

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II Industry Background

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Table 1: Market-Level Summary Statistics

	Mean	SD	p10	p90
Panel A: Market Characteristics				
Number of Products	24.3	8.1	14	35
Market Share (leader)	0.312	0.142	0.148	0.491
HHI	0.218	0.091	0.118	0.341
Panel B: Pricing				
Price ($)	24.81	12.41	10.50	41.20
Price-Cost Margin (%)	34.2	18.1	14.1	58.4
Markets	342	342	342	342
Products	8,321	8,321	8,321	8,321

Notes: Sample covers 342 geographic markets across 18 quarters. Price in constant 2010 dollars.

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Table 2: Product Characteristics Summary

	Mean	SD	Min	Max
Continuous Characteristics				
Quality Index (0-10)	6.42	1.84	1.20	9.91
Space (cu ft)	18.4	6.2	6.1	34.8
Energy Star Rating	3.21	1.14	1.00	5.00
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Notes: Product characteristics from manufacturer spec sheets matched to retail scanner data.

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III Demand Model

Consider an economy populated by a continuum of agents indexed by $i \in [0,1]$. Each agent chooses the outcome variable to maximize lifetime utility subject to a budget constraint. The key source of heterogeneity across agents is their exposure to the treatment variable. We solve for the equilibrium of this economy and derive comparative statics with respect to the key parameters. The model generates a clear testable prediction: an exogenous increase in the treatment variable should lead to a positive and significant change in the outcome variable, with the magnitude depending on structural parameters that we estimate in the empirical analysis.

Let Y_i denote the outcome of interest for unit i. We model Y_i as a function of treatment status D_i, observed covariates X_i, and unobserved heterogeneity ε_i. Under standard regularity conditions, the conditional expectation function takes the form presented in Equation **??**. The parameter of interest is the coefficient on treatment status, which captures the causal effect of the treatment variable on the outcome variable holding constant observed and unobserved confounders. Identification of this parameter requires an exclusion restriction, which we discuss in detail below.

The theoretical framework builds on the canonical model of field. Agents face a discrete choice between J alternatives, each yielding utility $U_{ij} = V_{ij} + \epsilon_{ij}$, where V_{ij} is the deterministic component and ϵ_{ij} captures idiosyncratic preferences. Under standard distributional assumptions on the error term, the choice probabilities take a tractable closed form. The model can be estimated using maximum likelihood or method of moments, and the resulting structural parameters can be used to simulate counterfactual policy scenarios.

We derive the key prediction of the model through a simple comparative statics exercise. An exogenous increase in the treatment variable shifts the equilibrium allocation of the outcome variable through two channels: a direct effect operating through prices, and an indirect effect operating through the reallocation of factors across sectors. The total effect is ambiguous in general, but under the empirically relevant parameter values, we show that the direct effect dominates. Proposition 1 in Appendix A formalizes this result and establishes conditions under

which the sign of the total effect is unambiguous.

Under the assumption that the proposed instrument satisfies the exclusion restriction, the model implies that the reduced-form relationship between the proposed instrument and the outcome variable is proportional to the structural parameter of interest. This motivates our instrumental variables strategy described in Section **??**. A key advantage of the IV approach is that it does not require functional form assumptions on the relationship between the treatment variable and the unobserved confounders. The cost is that we identify a local average treatment effect rather than the population average treatment effect.

$$s_{jt}(\delta, \Sigma) = \int \frac{\exp(\delta_{jt} + \mu_{ijt})}{1 + \sum_k \exp(\delta_{kt} + \mu_{ikt})} dP_\nu(\nu), \quad \Sigma = \begin{bmatrix} \sigma_p^2 & \rho_{pq}\sigma_p\sigma_q \\ \rho_{pq}\sigma_p\sigma_q & \sigma_q^2 \end{bmatrix} \tag{2}$$

$$\delta^{(r+1)} = \delta^{(r)} + \ln s^{obs} - \ln s(\delta^{(r)}, \Sigma), \quad \|\delta^{(r+1)} - \delta^*\| \leq \sqrt[3]{\binom{J}{2}\kappa^r \|\delta^{(0)} - \delta^*\|^3} \tag{3}$$

The model delivers additional testable implications beyond the main prediction. First, the treatment effect should be larger for individuals with lower baseline levels of the outcome variable, reflecting diminishing marginal returns. Second, the effect should be persistent over time rather than transitory, because the mechanism operates through permanent changes in the level of the treatment variable. Third, the effect should be concentrated in sectors with greater exposure to the underlying channel. We test each of these auxiliary predictions in Section 6.

We make several simplifying assumptions to maintain tractability. First, we assume that agents have rational expectations and perfect information about the distribution of shocks. Second, we abstract from general equilibrium effects by treating prices as exogenous to individual agents. Third, we assume that the production technology exhibits constant returns to scale. We discuss the robustness of our theoretical predictions to relaxing each of these assumptions in Appendix A, where we also present a more general version of the model.

The equilibrium of the model can be characterized by a system of first-order conditions and market-clearing conditions. In the interior solution, the marginal benefit of increasing the treatment variable equals the marginal cost, yielding an implicit equation for the optimal level of the

treatment variable as a function of exogenous parameters and state variables. We log-linearize this equation around the steady state to obtain the estimating equation that motivates our empirical specification.

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III.A Consumer Heterogeneity

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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III.B Market Share Inversion

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$$p - mc = -left(fracpartialspartialp'Omegaright)^{-1}s \tag{4}$$

$$p^m = mc + left(fracpartialspartialp'Omega^m right)^{-1}s(p^m) \tag{5}$$

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IV.A Firm Pricing

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We contribute to the literature on industrial organization by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in industrial organization is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors or-

thogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

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The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but

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IV.B Cost Estimation

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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$$\hat{theta} = argmin_{theta} xi(theta)' ZWZ' xi(theta) \tag{6}$$

$$Z^*_{jt} = Eleft[fracpartialxi_{jt} partialthetamidZright] = -Eleft[fracpartials_{jt} partialtheta/fracpartial \tag{7}$$

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A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

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V.A GMM Procedure

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V.B Instruments

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V.C Standard Errors

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VI Data

Our primary data source is administrative records, which covers 50,000 individuals over the period 1990–2010. This dataset provides detailed information on the outcome variable, the treatment variable, and a rich set of control variables that we use to address potential confounders. The data are available at the individual level and include both repeated cross-sections and a longitudinal component that allows us to track individuals over time. We discuss the advantages and limitations of this dataset relative to alternatives in the literature.

Table **??** presents summary statistics for the key variables used in our analysis. The mean of the outcome variable is consistent with values reported in prior studies, providing confidence that our sample is representative of the population of interest. The standard deviation suggests substantial variation across individuals, which is essential for our identification strategy. We also report the 10th and 90th percentiles to characterize the full distribution and to assess the extent to which our results might be driven by outliers in the tails.

We construct our main sample by merging several administrative and survey datasets. After applying standard sample restrictions and dropping observations with missing values on key

variables, our final estimation sample contains 50,000 observations covering 1990–2010. The merge rate across datasets exceeds 95 percent, alleviating concerns about selective attrition at the matching stage. We verify that the merged sample is representative by comparing moments of the key variables to those in the underlying source datasets.

The key independent variable, the treatment variable, exhibits substantial cross-sectional variation. The interquartile range spans a factor of three, providing sufficient variation to identify the effect of interest. We verify that this variation is not driven by outliers or measurement error by conducting a series of diagnostic tests. In particular, we show that the results are robust to winsorizing the treatment variable at the 1st and 99th percentiles, to using alternative measures from different data sources, and to correcting for measurement error using an errors-in-variables estimator.

We supplement the main dataset with auxiliary sources to construct additional control variables and instruments. Geographic controls come from standard sources in the literature, including distance to the coast, latitude, rainfall, temperature, and terrain ruggedness. Demographic controls are measured at baseline to avoid bad control problems. Institutional controls are drawn from widely used cross-country datasets. We lag all time-varying controls by one period to mitigate simultaneity concerns.

The measurement of the outcome variable deserves discussion. Following the standard approach in the literature, we use the logarithm of the outcome variable as our primary outcome variable. This transformation has three advantages: it reduces the influence of outliers, it allows us to interpret coefficients as approximate percentage changes, and it is consistent with the multiplicative structure of the theoretical model. We verify that our results are qualitatively robust to using the outcome variable in levels or in ranks.

Panel A of Table **??** reports summary statistics for the full sample, while Panels B and C restrict attention to subsamples that will be used in the heterogeneity analysis. The key variables are well-balanced across subsamples, with the exception of the treatment variable, which exhibits the expected variation driven by the quasi-experimental design. We formally test for balance across subsamples using a multivariate F-test and fail to reject the null of joint equality at

conventional significance levels.

Table 3: Market-Level Summary Statistics

	Mean	SD	p10	p90
Panel A: Market Characteristics				
Number of Products	24.3	8.1	14	35
Market Share (leader)	0.312	0.142	0.148	0.491
HHI	0.218	0.091	0.118	0.341
Panel B: Pricing				
Price ($)	24.81	12.41	10.50	41.20
Price-Cost Margin (%)	34.2	18.1	14.1	58.4
Markets	342	342	342	342
Products	8,321	8,321	8,321	8,321

Notes: Sample covers 342 geographic markets across 18 quarters. Price in constant 2010 dollars.

Table 4: Product Characteristics Summary

	Mean	SD	Min	Max
Continuous Characteristics				
Quality Index (0-10)	6.42	1.84	1.20	9.91
Space (cu ft)	18.4	6.2	6.1	34.8
Energy Star Rating	3.21	1.14	1.00	5.00
Products	8,321	8,321	8,321	8,321

Notes: Product characteristics from manufacturer spec sheets matched to retail scanner data.

Data quality is a potential concern in studies of this type. We address this concern in several ways. First, we cross-validate our primary measure of the outcome variable against alternative sources and find high concordance rates. Second, we examine the time series of key variables for breaks or anomalies that might indicate measurement problems. Third, we re-estimate our main specifications on a subsample that excludes observations flagged as potentially problematic by the data providers. All results are robust to these checks.

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VI.B Product Characteristics

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VII Results

Table **??** presents our main estimates. Column (1) reports the baseline specification without controls. Columns (2) through (4) progressively add demographic controls, geographic controls, and fixed effects. The point estimate remains remarkably stable across specifications, suggesting that selection on unobservables is unlikely to drive our results. Following the bounding approach of Oster (2019), we show that the degree of selection on unobservables required to explain away our estimate would need to be substantially larger than the degree of selection on observables.

The estimated coefficient on the treatment variable is statistically significant at the one percent level in all specifications. The magnitude implies that a one standard deviation increase in the treatment variable is associated with a positive and significant of approximately 0.3 standard deviations in the outcome variable. To put this in perspective, this effect size is comparable to the impact of an additional year of schooling on earnings, one of the most well-established causal relationships in economics. The 95 percent confidence interval is sufficiently narrow to rule out effects smaller than one-tenth of a standard deviation.

We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

The first-stage relationship between the proposed instrument and the treatment variable is strong, with an F-statistic well above conventional thresholds for weak instruments. The first-stage coefficient is precisely estimated and has the expected sign. Following the recommendations of Andrews, Stock, and Sun (2019), we also report the effective F-statistic and the tF critical values. Under all weak-instrument diagnostics considered, we can reject the null of a weak first stage at the 5 percent level.

$$varepsilon_{jjt} = fracpartials_{jt}partialp_{jt}cdotfracp_{jt}s_{jt} = fracp_{jt}s_{jt}intalpha_i s_{ijt}(1-s_{ijt})dP_{nu} \quad (8)$$

Heterogeneity analysis reveals that the treatment effect is concentrated among individuals with below-median baseline levels of the outcome variable. This pattern is consistent with the theoretical prediction that the marginal return to the treatment variable is decreasing. We also find suggestive evidence of heterogeneity along demographic dimensions, although the subsample estimates are less precisely estimated due to smaller sample sizes. The difference in treatment effects across subgroups is statistically significant at the 10 percent level.

We complement the regression analysis with non-parametric evidence on the treatment effect. Kernel density estimates of the distribution of the outcome variable show a clear rightward shift for treated individuals relative to untreated individuals. Quantile treatment effects, reported in the appendix, are positive and statistically significant throughout the distribution, with somewhat larger effects in the lower tail. This pattern suggests that the treatment reduces inequality in the outcome variable by disproportionately benefiting those at the bottom of the distribution.

The magnitude of our estimates is informative about the relative importance of the treatment variable as a determinant of the outcome variable. A variance decomposition exercise reveals that variation in the treatment variable can account for approximately 15 to 25 percent of the cross-sectional variance in the outcome variable, depending on the specification. While this leaves substantial residual variation unexplained, it establishes the treatment variable as one of the most quantitatively important determinants identified in the literature.

We also examine the dynamics of the treatment effect by interacting the treatment variable with indicators for different time horizons. The effect is small and statistically insignificant in the first year after treatment, grows steadily over the subsequent three years, and stabilizes at a new long-run level by year four. This gradual emergence of the treatment effect is consistent with the theoretical model, which predicts that the adjustment to the new equilibrium occurs over multiple periods as agents update their behavior and as general equilibrium effects propagate

Table 5: Demand Estimates: Simple Logit

	OLS	IV	IV	IV
Dep. var.: Log Market Share Ratio				
Price	-0.041***	-0.182***	-0.171***	-0.164***
	((0.006))	((0.031))	((0.029))	((0.027))
Quality	0.318***	0.341***	0.328***	0.319***
	((0.028))	((0.031))	((0.029))	((0.027))
Space	0.024***	0.027***	0.025***	0.024***
	((0.004))	((0.005))	((0.004))	((0.004))
Energy Star	0.112***	0.118***	0.114***	0.111***
	((0.019))	((0.021))	((0.020))	((0.018))
Market FE	Yes	Yes	Yes	Yes
Quarter FE	Yes	Yes	Yes	Yes
Brand FE	No	No	Yes	Yes
Observations	149,778	149,778	149,778	149,778
R-squared	0.741			

Notes: *** p<0.01. Instruments: BLP instruments (sum of other products' characteristics). Market-clustered SEs.

Table 6: Demand Estimates: Random Coefficients Logit (BLP)

	Mean $\bar{\beta}$	SD σ	t-stat ($\bar{\beta}$)	t-stat (σ)
Random Coefficients Logit Estimates				
Price	-0.241	0.118	-7.82	4.21
	((0.031))	((0.028))		
Quality	0.384	0.142	12.41	5.81
	((0.031))	((0.024))		
Space	0.031	0.014	7.41	3.62
	((0.004))	((0.004))		
Energy Star	0.128	0.041	6.12	2.91
	((0.021))	((0.014))		
Observations	149,778	149,778	149,778	149,778
GMM Objective	18.42	18.42	18.42	18.42
Overid. p-value	0.318	0.318	0.318	0.318

Notes: BLP random coefficients logit. Instruments: BLP + Hausman price instruments. Markets pooled.

Table 7: Own-Price Elasticities by Segment

	Logit	RC Logit	Nested Logit	RC Nested
Mean Own-Price Elasticity by Segment				
Budget Segment	-3.21	-4.18	-3.84	-4.41
	((0.18))	((0.31))	((0.24))	((0.38))
Mid Segment	-2.84	-3.61	-3.42	-3.88
	((0.16))	((0.27))	((0.21))	((0.34))
Premium Segment	-2.41	-3.14	-2.98	-3.31
	((0.14))	((0.24))	((0.19))	((0.31))
All Products	-2.91	-3.71	-3.42	-3.94
	((0.17))	((0.28))	((0.22))	((0.35))

Notes: Elasticities evaluated at sample means. Bootstrap standard errors in parentheses.

Table 8: Cross-Price Elasticities: Selected Products

	Product A	Product B	Product C	Product D
Cross-price elasticity matrix (RC Logit)				
Product A price	-3.84	0.028	0.019	0.011
	((0.31))	((0.004))	((0.003))	((0.002))
Product B price	0.031	-3.61	0.024	0.014
	((0.004))	((0.27))	((0.003))	((0.002))
Product C price	0.021	0.026	-3.42	0.018
	((0.003))	((0.004))	((0.28))	((0.003))
Product D price	0.012	0.015	0.019	-3.14
	((0.002))	((0.002))	((0.003))	((0.28))

Notes: RC Logit estimates. Own-price elasticities on diagonal; cross-price elasticities off diagonal.

through the economy.

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VII.A Demand Estimates

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We find no evidence that our results are driven by pre-existing trends or differential selection. The event study coefficients are small and statistically insignificant in all pre-treatment periods, lending support to our parallel trends assumption. Moreover, the treatment effect emerges sharply at the time of the intervention rather than gradually, which is consistent with a causal interpretation and inconsistent with pre-existing trends that happen to accelerate at the time of treatment.

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VII.B Elasticities

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on industrial organization by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association,

while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in industrial organization is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification. This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the

extensive body of work on the determinants of the outcome variable, which has been a central topic in industrial organization since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to approximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

VII.C Markups

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

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Table 9: Estimated Markups

	Logit	RC Logit	Nested	Data (Acctg)
Price-Cost Markup (%)				
Budget Segment	21.4	28.1	24.8	25.2
	((1.2))	((2.1))	((1.8))	((3.1))
Mid Segment	26.8	33.4	29.8	30.4
	((1.4))	((2.4))	((2.0))	((3.4))
Premium Segment	31.2	38.4	34.1	36.8
	((1.8))	((2.8))	((2.4))	((3.8))

Notes: RC Logit markups closest to accounting-based estimates. Bootstrap SEs in parentheses.

cussion of policy implications.

Our paper relates to several strands of the economics literature. First, we contribute to the extensive body of work on the determinants of the outcome variable, which has been a central topic in industrial organization since the seminal contributions of the 1990s and 2000s. Second, we contribute to the methodological literature on an instrumental variables strategy by demonstrating its applicability to a new empirical context. Third, our findings speak to an ongoing policy debate about the effectiveness of interventions targeting the treatment variable.

VIII Counterfactuals

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on industrial organization by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assump-

tion underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

The existing literature on the relationship between the treatment variable and the outcome variable has produced mixed results. Some studies find a positive and significant association, while others find no statistically significant effect. We argue that these discrepancies arise from differences in identification strategies and sample composition. Studies relying on cross-sectional variation tend to overstate the relationship due to omitted variable bias, while panel data approaches that rely on within-unit variation may attenuate the true effect by absorbing long-run changes in the treatment variable. Our approach navigates between these extremes by exploiting quasi-experimental variation.

A central question in industrial organization is whether changes in the treatment variable lead to changes in the outcome variable, or whether the observed correlation reflects reverse causality or omitted variable bias. We address this question using an instrumental variables strategy applied to data from administrative records. The institutional setting we study provides a compelling natural experiment because the relevant policy changes were driven by factors orthogonal to contemporaneous economic conditions. We document this independence using a battery of balance tests and falsification exercises.

$$CV_i = -frac1alpha_i lnleft(sum_j exp(u_{ij}^{post})right) + frac1alpha_i lnleft(sum_j exp(u_{ij}^{pre})right) \tag{9}$$

Our main finding is that the treatment variable has a statistically significant and economically meaningful effect on the outcome variable. The magnitude of this effect is robust across a wide range of specifications, subsamples, and alternative measures of the key variables. A one standard deviation increase in the treatment variable is associated with a change of approximately 0.3 to 0.5 standard deviations in the outcome variable, depending on the specification.

This magnitude is consistent with the predictions of the theoretical model developed in Section 2 and is comparable to estimates from related quasi-experimental studies.

The remainder of this paper proceeds as follows. Section 2 presents the theoretical framework that motivates our empirical analysis and generates testable predictions. Section 3 describes the institutional background and the natural experiment we exploit. Section 4 describes the data sources and sample construction. Section 5 outlines our empirical strategy, including the identification assumptions and potential threats to validity. Section 6 presents the main results. Section 7 discusses robustness checks and extensions. Section 8 concludes with a discussion of policy implications.

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Table 10: Merger Simulation: Price Effects

	Logit	RC Logit	Nested	RC Nested
Predicted Price Change from Merger (%)				
Merging Firm A Products	4.12	6.84	5.91	7.21
	((0.38))	((0.71))	((0.54))	((0.82))
Merging Firm B Products	3.84	6.41	5.48	6.78
	((0.34))	((0.64))	((0.49))	((0.74))
Rival Products	1.21	1.94	1.68	2.11
	((0.14))	((0.24))	((0.19))	((0.28))

Notes: Bertrand-Nash merger simulation. RC Logit predicts larger price increases than simple logit.

The economic significance of our findings warrants emphasis. Back-of-the-envelope calculations suggest that the estimated effect implies substantial welfare gains from policies that improve the treatment variable. Using the framework of Chetty (2009) for translating reduced-form estimates into welfare statements, we estimate that a policy intervention that increases the treatment variable by one standard deviation would generate welfare gains equivalent to ap-

Table 11: Welfare Effects of Proposed Merger

	Logit	RC Logit	Nested	RC Nested
Welfare Change ($ per consumer per quarter)				
Consumer Surplus Change	-1.84	-3.12	-2.68	-3.41
	((0.21))	((0.38))	((0.31))	((0.44))
Producer Surplus Change	2.41	3.84	3.21	4.12
	((0.28))	((0.44))	((0.38))	((0.51))
Total Welfare Change	0.57	0.72	0.53	0.71
	((0.14))	((0.22))	((0.18))	((0.26))

Notes: Compensating variation consumer surplus measure. RC Logit suggests larger welfare effects.

proximately 2-4 percent of annual income. These calculations are necessarily approximate but suggest that the magnitudes involved are economically meaningful.

A distinctive feature of our analysis is the richness of the data. Unlike many previous studies that rely on aggregate proxies, our dataset contains individual-level information on both the outcome variable and the treatment variable, measured with high frequency over an extended time period. This granularity allows us to investigate heterogeneity in treatment effects across individuals, to construct precise event study estimates, and to conduct a range of placebo and falsification tests that would not be feasible with more aggregated data.

Before proceeding, we note several contributions relative to the closest existing work. While prior studies have documented a correlation between the treatment variable and the outcome variable in various contexts, our paper is among the first to provide credibly causal estimates using an instrumental variables strategy. Moreover, our analysis of mechanisms goes beyond what has been possible in earlier studies, shedding light on the channels through which the treatment variable affects the outcome variable.

VIII.A Merger Simulation

This paper investigates the causal relationship between the treatment variable and the outcome variable. Understanding this relationship is critical for both economic theory and policy. Despite a growing literature, credible identification of the causal effect remains challenging due to potential confounders and reverse causality. We contribute to this literature by proposing a new

identification strategy based on an instrumental variables strategy. Our approach addresses the key threats to internal validity that have limited the conclusions of prior work.

We contribute to the literature on industrial organization by exploiting a novel source of variation in the treatment variable. Our identification strategy relies on an instrumental variables strategy, which allows us to isolate the causal effect of the treatment variable on the outcome variable from confounding factors that have plagued earlier studies. The identifying assumption underlying our approach is that the proposed instrument affects the outcome variable only through its effect on the treatment variable, conditional on observable characteristics. We provide several pieces of evidence in support of this assumption, including placebo tests, overidentification tests, and bounds analysis.

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VIII.B Welfare Analysis

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IX Conclusion

This paper provides new evidence on the causal effect of the treatment variable on the outcome variable. Using an instrumental variables strategy applied to administrative records, we find a positive and significant and statistically significant effect that is robust across a wide range of specifications. The magnitude of the effect is economically meaningful and has clear implications for both theory and policy.

Our findings have several implications for economic theory. First, the results support models in which the treatment variable plays a central role in determining the outcome variable, as opposed to alternative theories that emphasize other factors. Second, the pattern of heterogeneity in treatment effects is consistent with the predictions of the theoretical framework, providing a rare example of a model that is both internally consistent and empirically validated. Third, the dynamics of the treatment effect shed light on the speed of adjustment to economic shocks, a parameter that is notoriously difficult to estimate.

From a policy perspective, the estimates suggest that interventions targeting the treatment variable could have meaningful effects on the outcome variable. However, several caveats are in order. First, our estimates are local to the population and context studied, and external validity to other settings should not be assumed without additional evidence. Second, our identification strategy relies on assumptions that, while supported by the evidence, cannot be directly tested. Third, the general equilibrium effects of large-scale policy interventions may differ from the partial equilibrium effects captured by our estimates.

Future work should investigate the mechanisms through which the treatment variable affects the outcome variable in greater detail. Our results identify a reduced-form relationship but do not fully pin down the structural channels at work. Complementary evidence from randomized controlled trials, structural estimation, or natural experiments in other contexts would strengthen the conclusions drawn here. Additionally, understanding the long-run persistence of the effects documented in this paper is an important question for future research.

In conclusion, this paper demonstrates that the treatment variable is a quantitatively im-

portant determinant of the outcome variable. The combination of a compelling identification strategy, rich data, and robust results across specifications provides a strong foundation for the causal claims made here. We hope that these findings will stimulate further research on the economic consequences of variation in the treatment variable and will inform the design of policies aimed at improving the outcome variable.

This paper provides new evidence on the causal effect of the treatment variable on the outcome variable. Using an instrumental variables strategy applied to administrative records, we find a positive and significant and statistically significant effect that is robust across a wide range of specifications. The magnitude of the effect is economically meaningful and has clear implications for both theory and policy.

Our findings have several implications for economic theory. First, the results support models in which the treatment variable plays a central role in determining the outcome variable, as opposed to alternative theories that emphasize other factors. Second, the pattern of heterogeneity in treatment effects is consistent with the predictions of the theoretical framework, providing a rare example of a model that is both internally consistent and empirically validated. Third, the dynamics of the treatment effect shed light on the speed of adjustment to economic shocks, a parameter that is notoriously difficult to estimate.

From a policy perspective, the estimates suggest that interventions targeting the treatment variable could have meaningful effects on the outcome variable. However, several caveats are in order. First, our estimates are local to the population and context studied, and external validity to other settings should not be assumed without additional evidence. Second, our identification strategy relies on assumptions that, while supported by the evidence, cannot be directly tested. Third, the general equilibrium effects of large-scale policy interventions may differ from the partial equilibrium effects captured by our estimates.

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In conclusion, this paper demonstrates that the treatment variable is a quantitatively important determinant of the outcome variable. The combination of a compelling identification strategy, rich data, and robust results across specifications provides a strong foundation for the causal claims made here. We hope that these findings will stimulate further research on the economic consequences of variation in the treatment variable and will inform the design of policies aimed at improving the outcome variable.

X Appendix A: Proofs

This appendix provides additional results referenced in the main text. We organize the supplementary material into three parts. Appendix A presents proofs of the theoretical propositions stated in Section **??**. Appendix B reports additional empirical results, including alternative specifications and sample definitions. Appendix C describes the construction of key variables in detail.

We present detailed derivations of the key theoretical results stated in Section **??**. The proofs proceed under standard regularity conditions that are verified in the data. We state these conditions formally as Assumptions A1 through A4 below, and provide discussion of their economic content and empirical plausibility. Throughout, we maintain the notation introduced in the main text.

This section reports results from additional robustness exercises not included in the main text due to space constraints. All estimates are qualitatively similar to the baseline results reported in Table **??**. We organize these results by the type of robustness check: alternative control variables, alternative sample restrictions, alternative functional forms, alternative estimation methods, and alternative definitions of key variables.

Proposition 1 (Contraction Mapping: Berry 1994)**.** *Define the operator $T : \mathbb{R}^J \to \mathbb{R}^J$ by*

$$T(\delta)_j = \delta_j + \ln s_j^{obs} - \ln s_j(\delta, \Sigma). \tag{10}$$

T is a contraction mapping with modulus $\kappa < 1$, so the iteration $\delta^{(r+1)} = T(\delta^{(r)})$ converges to the unique fixed point $\delta^* = \sigma^{-1}(s^{obs}, \Sigma)$.

Proof. The Jacobian satisfies

$$\frac{\partial T_j}{\partial \delta_k} = \begin{cases} 1 - s_j & k = j \\ s_k & k \neq j. \end{cases} \tag{11}$$

Since $\sum_k \partial s_j / \partial \delta_k = s_j(1 - s_j) - \sum_{k \neq j} s_j s_k = 0$, the operator norm $\| I - \partial T / \partial \delta \|_\infty < 1$. For mixed logit, dominated convergence extends this bound uniformly over ν. □

Proposition 2 (GMM Consistency)**.** *Under compactness of Θ, identification, and $E[\xi_{jt} z_{jt}] = 0$, the BLP GMM estimator*

$$\hat{\theta} = \arg\min_{\theta} \xi(\theta)' Z W Z' \xi(\theta) \tag{12}$$

satisfies $\hat{\theta} \xrightarrow{p} \theta_0$ as $T \to \infty$.

Proposition 3 (Optimal Instruments)**.** *By the Chamberlain (1987) efficiency bound, the optimal instrument for θ_k is*

$$Z_k^* = E\left[\frac{\partial \xi}{\partial \theta_k} \mid p, x, Z_{exog} \right] = E\left[\left(\frac{\partial s}{\partial \delta} \right)^{-1} \frac{\partial s}{\partial \theta_k} \mid Z_{exog} \right]. \tag{13}$$

Proposition 4 (Bertrand-Nash Equilibrium Prices)**.** *Under demand $s(p)$ with $\partial s_j / \partial p_j < 0$, $|\partial s_j / \partial p_j| > |\partial s_k / \partial p_j|$ for $k \neq j$, and convex costs, there exists a unique equilibrium price vector p^* satisfying*

$$p_j^* - mc_j = -\frac{s_j(p^*)}{\partial s_j / \partial p_j(p^*)}, \quad \forall j \in \mathscr{F}_f, \tag{14}$$

for each firm f's product portfolio $\mathscr{F}_f$.

Proof. Define $\phi(p) = mc - (\partial s/\partial p)^{-1}\Omega^{-1}s(p)$. Existence follows from Brouwer's fixed point theorem on $[mc, \bar{p}]^J$. Uniqueness follows from the P-matrix property of $\partial s/\partial p$ under logit demand. □

Combinatorial structure of instruments. There are $\binom{J}{2}$ distinct product-pair interactions in the BLP instrument set. The optimal instrument matrix has the block structure

$$Z^* = \begin{bmatrix} z_{11}^* & z_{12}^* & \cdots \\ z_{21}^* & z_{22}^* & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}, \quad \sqrt[3]{\det(Z^{*\prime}Z^*)} \leq \prod_{k=1}^{K} \|z_k^*\|^{2/K}. \tag{15}$$

The convergence rate of the contraction mapping satisfies $\|\delta^{(r)} - \delta^*\| \leq \sqrt[3]{\kappa^r} \cdot \|\delta^{(0)} - \delta^*\|$.

We provide here a detailed description of the data sources and variable construction procedures. For each variable, we report the original source, the years available, the unit of observation, and any transformations applied. We also report summary statistics for the full set of variables used in the analysis, including those that appear only in robustness checks or auxiliary regressions.

XI Appendix B: Computational Details

This appendix provides additional results referenced in the main text. We organize the supplementary material into three parts. Appendix A presents proofs of the theoretical propositions stated in Section **??**. Appendix B reports additional empirical results, including alternative specifications and sample definitions. Appendix C describes the construction of key variables in detail.

We present detailed derivations of the key theoretical results stated in Section **??**. The proofs proceed under standard regularity conditions that are verified in the data. We state these conditions formally as Assumptions A1 through A4 below, and provide discussion of their economic

content and empirical plausibility. Throughout, we maintain the notation introduced in the main text.

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The following tables report the complete set of coefficient estimates for all control variables included in our main specifications. In the interest of space, the main text reports only the coefficients on the key independent variables. The full results presented here confirm that the control variables enter with expected signs and magnitudes, and that their inclusion does not materially affect the coefficients of interest.

Table 12: Appendix: Instrument Relevance and Validity

	F-stat	Partial R2	Hansen J	p-value
Instrument Diagnostics				
BLP Instruments (own chars)	84.2	0.182	12.41	0.412
BLP Instruments (rival chars)	91.4	0.201	11.84	0.458
Hausman Instruments	72.1	0.161	13.12	0.361
Optimal Instruments	142.8	0.318	8.41	0.681

Notes: Optimal instruments constructed from nonparametric estimates of $E[\partial s/\partial \theta]$.

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XII Appendix C: Monte Carlo Design

This appendix provides additional results referenced in the main text. We organize the supplementary material into three parts. Appendix A presents proofs of the theoretical propositions stated in Section **??**. Appendix B reports additional empirical results, including alternative specifications and sample definitions. Appendix C describes the construction of key variables in detail.

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Table 13: Monte Carlo: Bias of Estimators

	N=200	N=500	N=1000	N=5000
Bias in Price Coefficient ($\times 100$)				
OLS	41.2	40.8	41.1	40.9
	((2.1))	((1.4))	((1.0))	((0.4))
BLP (BLP Instruments)	1.8	1.2	0.8	0.3
	((1.9))	((1.3))	((0.9))	((0.4))
BLP (Optimal Instruments)	0.9	0.6	0.4	0.1
	((1.8))	((1.2))	((0.8))	((0.3))

Notes: 1000 Monte Carlo replications. True $\bar{\beta}_p = -0.241$, $\sigma_p = 0.118$.

Table 14: Monte Carlo: Coverage Rates of 95% Confidence Intervals

	N=200	N=500	N=1000	N=5000
Coverage Rate (%)				
BLP (BLP Instruments)	88.4	91.2	93.1	94.8
	((1.0))	((0.9))	((0.8))	((0.7))
BLP (Optimal Instruments)	91.2	93.4	94.2	94.9
	((0.9))	((0.8))	((0.7))	((0.7))

Notes: Nominal coverage rate: 95%. BLP instruments achieve near-nominal coverage at $N \geq 1000$.

cients on the key independent variables. The full results presented here confirm that the control variables enter with expected signs and magnitudes, and that their inclusion does not materially affect the coefficients of interest.

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